AI CAREERS FOR WOMEN

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SmartEdu: Optimizing Institutional Resource Allocation for Maximum Academic ROI

Project Domain

Excel / Power BI / Power BI with Copilot / AI Chatbot

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3. Problem Statement

A. Background of Educational Resource Allocation Educational institutions invest heavily in various academic resources—such as hiring high-quality teachers, funding tutoring programs, and upgrading infrastructure. Despite these investments, administrators lack a data-driven framework to measure the actual academic "Return on Investment" (ROI) generated by these specific initiatives.

B. The Financial and Operational Inefficiency Currently, institutional capital is often misallocated using a "one-size-fits-all" approach. Schools struggle to determine whether a limited budget is better spent on high-cost academic interventions or low-cost environmental changes. This lack of precision leads to budget inefficiencies.

C. The Accessibility and Interpretation Gap Even when educational data is analysed, the findings are usually locked in static spreadsheets or dense reports. School administrators and counsellors are not data scientists; they need a way to interact with the data naturally and ask immediate, context-specific questions (e.g., "Which students need tutoring the most this week?") without navigating complex databases.

D. The Analytical Need The core problem is the absence of a dynamic, predictive model combined with an intuitive user interface. There is an immediate need to translate raw student data into an interactive visual dashboard and a conversational AI assistant, allowing institutions to strategically deploy their budgets and intervene in real-time.

4. Objectives

Calculate Academic ROI: Quantify the direct performance impact of high-cost interventions on final exam scores to determine their actual financial and academic value.

Develop a Segmented Investment Strategy: Analyse how external socio-economic factors influence academic outcomes, allowing for targeted, demographic-specific interventions.

Deploy Advanced Business Intelligence: Build a dynamic Power BI dashboard that connects live data models, enabling stakeholders to drill down into specific school types, income brackets, and performance metrics.

Implement Conversational AI Analytics: Develop an interactive Chatbot that allows school administrators to query the dataset using natural language to retrieve instant insights on student risk factors and resource allocation recommendations.

5. Methodology / Implementation Plan

Phase	Task Description	Tools / Techniques	Tentative Timeline
1. Data Structuring & Modelling	Clean and standardize the dataset of 7,500+ records. Create a relational data model mapping "Institutional Investments" against "Student Behaviours" and "Socio-Economic Baselines."	Excel Power Query, Data Modelling	Week 1
2. Statistical & ROI Analysis	Calculate the marginal gain in exam scores when altering variables. Run correlation and regression analysis to isolate the true drivers of performance.	Excel Analysis ToolPak	Week 2
3. Advanced Visualization	Import the structured data model into Power BI. Develop interactive visualizations, DAX measures for ROI tracking, and drill-through pages for specific demographic analysis.	Power BI Desktop, DAX	Week 3
4. AI Chatbot Integration	Build an interactive Chatbot configured to read the dataset/dashboard. Train the bot to answer administrative queries like "Show me the pass rate for low-income students with high tutoring."	Microsoft Copilot Studio / Power BI Q&A feature	Week 3 - 4
5. Testing & Final	Test the dashboard interactions and chatbot accuracy. Compile the final	Power BI Service, GitHub,	Week 4

Phase	Task Description	Tools / Techniques	Tentative Timeline
Reporting	analytical report and prepare the project for presentation.	Word	

B. Dataset

The dataset used for this project is titled **"Student Performance Factors"**, containing approximately **6500 student records** and **20 variables** related to academic performance, behavioral traits, and socio-economic background.

The dataset consists of the following types of variables:

Numerical Variables (Continuous/Integer):
Hours Studied, Attendance, Sleep Hours, Previous Scores, Tutoring Sessions, Physical Activity, Exam Score

Categorical Variables (Text/Ordinal):
Gender, School Type, Family Income, Motivation Level, Teacher Quality, Parental Involvement, Peer Influence, Parental Education Level, Access to Resources

Binary Variables (Yes/No):
Internet Access, Extracurricular Activities, Learning Disabilities

The target variable for analysis is **Exam Score**, which is used to evaluate academic performance and classify students into grade categories.

Source of Dataset

The dataset has been sourced from a publicly available academic dataset repository (Kaggle). https://www.kaggle.com/datasets/lainguyn123/student-performance-factors/data

It is suitable for exploratory data analysis and performance modeling in the education domain.

7. Expected Outcomes / Conclusion

An Interactive Power BI Dashboard: A highly visual, dynamic tool replacing static reports, allowing school boards to filter and track the effectiveness of educational inputs across thousands of student records.

An "Edu-Advisor" Chatbot: A functional conversational AI interface where administrators can type natural language questions and instantly receive data-backed answers and charts regarding student performance and budget allocation.

An Investment Prioritization Matrix: A comprehensive report detailing which factors institutions should prioritize funding first to achieve the highest academic yield.

Demonstrated Full-Stack BI Skills: A polished portfolio piece showcasing the complete data lifecycle: from raw Excel data cleaning to advanced Power BI visualization and modern AI Chatbot integration.

8. References

Microsoft Corporation. (2024). *Power BI Documentation and DAX Reference*. Available at: https://learn.microsoft.com/en-us/power-bi/

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Kaggle Inc. *Open Datasets - Student Performance Factors*. Available at: https://www.kaggle.com/datasets

Levin, H. M., & Belfield, C. (2015). *Guiding the Development and Use of Cost-Effectiveness Analysis in Education*. Journal of Research on Educational Effectiveness.